

Sir Syed University of Engineering & Technology
Department of Computer Science & Information Technology
Computer Organization & Assembly Language (CS-215T)

— ARITHMETIC AND LOGIC INSTRUCTIONS & PROGRAMS —

Unsigned Addition And Subtraction

ADDITION:

ADD destination, source ; *Half Adder*
ADC destination, source ; *Full Adder*

CASE 1: *Addition of individual byte and word data*

CASE 2: *Addition of multiword numbers*

ADD without carry

```
BACK: ADD AL, [SI]
      JNC OVER
      INC AH
OVER:  INC SI
```

ADD with carry

```
BACK: ADD AL, [SI]
      ADC AH, 00
      INC SI
```

SUBTRACTION:

SUB destination, source ; *Half Subtractor*
SBB destination, source ; *Full Subtractor*

"SUB" Step:

1. 2's complement of source
2. Add result to destination
3. Invert carry

3a. $CF = 1 \Rightarrow$ Signed Number (-ve result)
3b. $CF = 0 \Rightarrow$ Positive Number

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UNSIGNED MULTIPLICATION & DIVISION

MULTIPLICATION:

syntax is:

MUL operand2 ; AX holds operand1.

SUMMARY:

MULTIPLICATION	OP1	OP2	RESULT
byte x byte	AL	Register Or Memory	AX
word x word	AX	Register Or Memory	DX: AX
word x byte	{ AL=byte AH=0	Register Or Memory	DX: AX

DIVISION:

syntax is:

DIV denominator ; AX holds Numerator

SUMMARY:

DIVISION	NUMERATOR	DENOMINATOR	Q	RESULT	R
byte/byte	AL=byte ; AH=0	Reg. Or Mem.	AL		AH
word/word	AX=word ; DX=0	Reg. Or Mem.	AX		DX
word/byte	AX=word	Reg. Or Mem.	AL		AH
DW/word	DX:AX = DW	Reg. Or Mem.	AX		DX

Divide Error Condition:

- AL > FFH
- AX > FFFFH

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ADD 2 Quad Words.
 $548FB9963CE7_H + 3FCD4FA23B8D_H$
 at 0000_H → ← at 0010_H
 RESULT → at 0020_H

Memory status:

00	01	02	03	04	05	06	07	...	0F
E7	3C	96	89	8F	54	00	00		

10	11	12	13	14	15	16	17	...	1F
8D	3B	A2	4F	CD	3F	00	00		

20	21	22	23	24	25	26	27	...	2F
—	—	—	—	—	—	—	—		

Requirements:

CX

BX

AX

SI

DI

Looping

Adding with carry i.e.; ADC

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ADDING MULTIWORD NUMBERS

- MODEL SMALL
- STACK 64
- DATA

```
DATA1 DQ 548FB9963CE7H
      ORG 0010H
DATA2 DQ 3FCD4FA23B8DH
      ORG 0020H
DATA3 DQ 7
```

- CODE

```
MAIN: MOV AX, @DATA
      MOV DS, AX
      CLC
      MOV SI, OFFSET DATA1
      MOV DI, OFFSET DATA2
      MOV BX, OFFSET DATA3
      MOV CX, 04

BACK: MOV AX, [SI]
      MOV ADC AX, [DI]
      MOV [BX], AX
      INC SI
      INC SI
      INC DI
      INC DI
      INC BX
      INC BX
      LOOP BACK
      MOV AH, 4CH
      INT 21H

END MAIN
```


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— ARITHMETIC AND LOGIC INSTRUCTIONS & PROGRAMMING —

LOGIC INSTRUCTIONS:

➤ **AND DESTINATION, SOURCE**
 USE: 1. TESTING ZERO OPERAND
 2. MASKING

$\begin{cases} CF = 0 \\ OF = 0 \end{cases}$
 PF = ZF = SF = XX

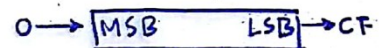
➤ **OR DESTINATION, SOURCE**
 USE: 1. TESTING ZERO OPERAND
 2. SETTING particular BIT/S.

$\begin{cases} CF = 0 \\ OF = 0 \end{cases}$

➤ **XOR DESTINATION, SOURCE**
 USE: 1. Toggling of BITS

$\begin{cases} CF = 0 \\ OF = 0 \end{cases}$

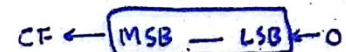
➤ **SHR destination, count**



NOTE:

- destination can be Register or Memory
- if count > 1 then CL = count
 else if count == 1 then count = 1

Similarly ➤ **SHL destination, count**



⚠ destination can't be an IMMEDIATE VALUE

➤ **CMP destination, source**

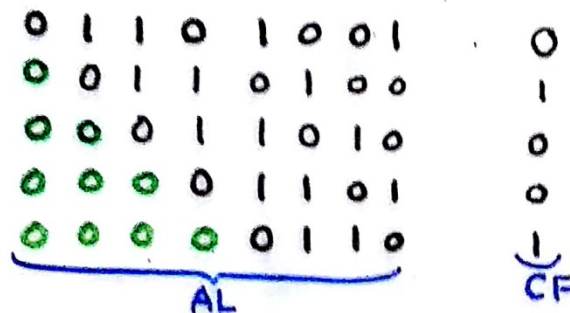
↙
 Register
 Memory

↘
 Register
 Memory
 Immediate

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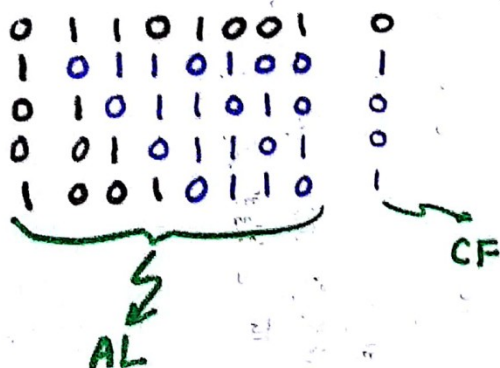
SHIFTING BITS 'RIGHT OR LEFT'

SHR: CLC
 MOV AL, 01101001B
 MOV CL, 4
 SHR AL, CL



ROTATING BITS 'WITH & WITHOUT CARRY'

ROR: CLC
 MOV AL, 01101001B
 MOV CL, 4
 ROR AL, CL



RCL: CLC
 MOV AL, 01101001B
 MOV CL, 4
 RCL AL, CL

